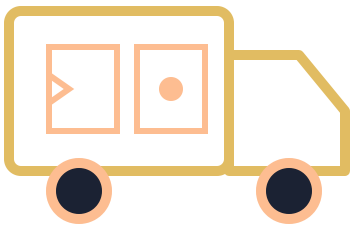
TEACHING CASE · GROCERY · SUPPLY CHAIN

When the delivery truck stops. Check the files at the door.

How a three-store grocer and its smart-cart app, Glidecart, can stop tampered supplier files and keep fresh food on the shelves when a distributor goes dark.



20 of 20

tampered supplier files stopped by the new gateway

26 → 2.5 h

to place an emergency perishable order in an outage

430 hrs

of invoice-checking time saved each year

Anèl Henning · Logical Coders, Tampa, FL

Fictional organization · illustrative data

01 · THE WEAK SPOT

A grocery store runs on files it cannot see

NOV 2024
Ahold Delhaize USA, parent of Food Lion and Hannaford, takes systems offline to contain a cyber incident (Ahold Delhaize, 2024).

DEC 2024
Attackers exploit Cleo file-transfer software, which many companies use to swap files with partners (CISA, 2024).

JUN 2025
A cyberattack on distributor UNFI disrupts its ability to fill and ship store orders (United Natural Foods, Inc., 2025).

THIS CASE
A small grocer controls the door its files come through, and practices for the day the truck stops.

3	45	$27M	240	0
STORES + DELI	ASSOCIATES	YEARLY SALES	INVOICES / MONTH	OUTAGE PLANS

Milk, produce, and deli meat have short shelf lives, and every order depends on files moving between a store and its suppliers. When those files stop, or when someone changes them, the store loses money fast.

A small grocer cannot prevent attacks on its suppliers. It can control how it receives files and how it keeps working when a supplier goes down. NIST's supply chain guidance makes the same point: manage the risk at the connection points you control (Boyens et al., 2022).

Bayside Fresh Market is a fictional three-store grocer. It receives invoices and price files from one main distributor through an older file-transfer server that faces the internet. Glidecart, its fictional smart-cart and pricing app, reads the daily price file so carts show the right prices, and it hosts the portal where suppliers upload invoices.

ABOUT THIS CASE

Bayside Fresh Market and Glidecart's role here are fictional, and every trial number is illustrative. The real incidents are cited from public sources.

STARTING AT THE DELI COUNTER

Using the ISO 9241-210 cycle (ISO, 2019), we interviewed the deli lead, two store managers, the accounts payable (AP) clerk, and the Glidecart developer. Security should fit how people actually work (Sasse et al., 2001).

- **Deli lead:** "If the truck doesn't come, I don't know who to call or what to order."
- **AP clerk:** "I can't tell a fake invoice from a real one. They all look the same."
- **Define.** Both are design problems, not training failures. The fix must fit an AP clerk's morning and a deli lead's shift.

02 · WHAT WE SET OUT TO PROVE

Three hypotheses, three trials

H1 · SECURITY

A gateway catches tampered files

Known-sender, fingerprint, and format checks, plus a call-back on any bank-detail change, will stop more tampered files than today's process.

SUPPORTED

H2 · CONTINUITY

A practiced fallback saves the shelves

A rehearsed paper and phone plan, the "Truck Stop Binder," will shorten the time to place an emergency perishable order.

SUPPORTED

H3 · LEARNING

Hands-on sorting builds skill

Associates trained with a tactile board and the dashboard will sort files more accurately and feel more capable than those given a memo.

STRONG TREND

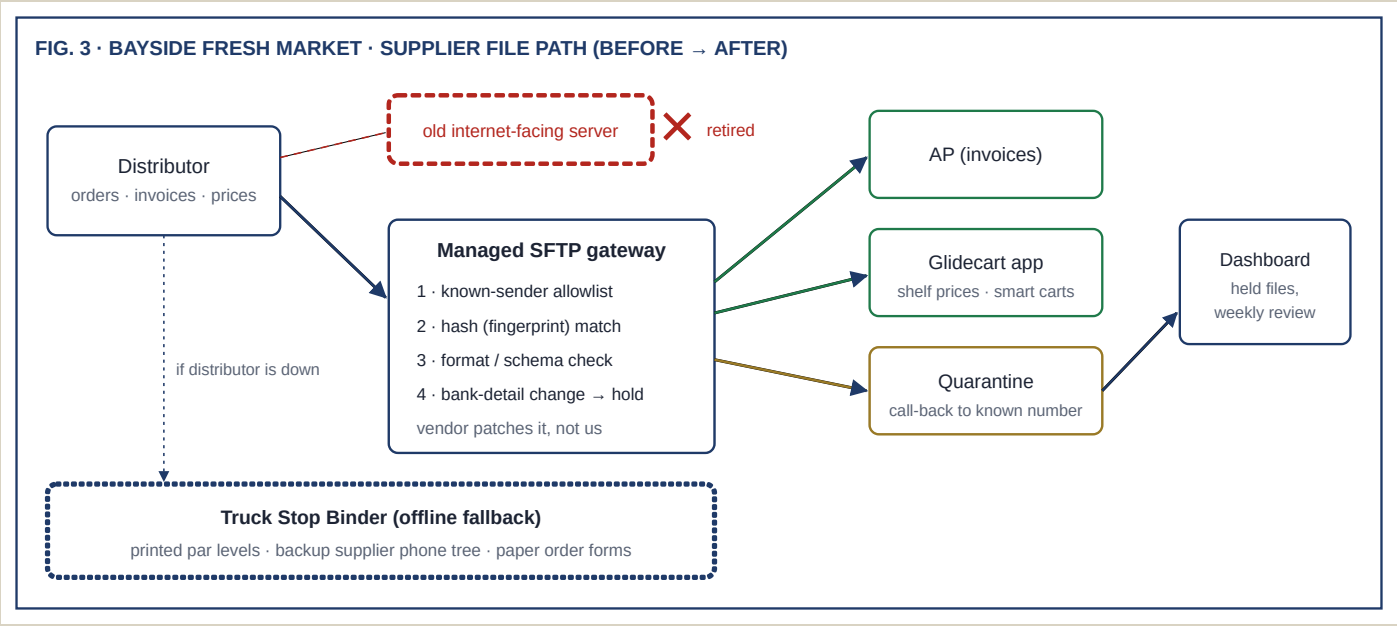


Figure 1. How the solution is built. The old internet-facing server is retired. Supplier files go through a managed gateway that checks the sender, the file fingerprint, and the format, and holds any bank-detail change for a call-back. Clean files flow to AP and Glidecart. If the distributor goes offline, stores switch to the printed Truck Stop Binder.

TRIAL A · TAMPERED-FILE TEST

Over eight weeks, 40 test files went through each path: the current server with a manual AP check, and the new gateway. Twenty in each set were tampered with, by changing bank details, altering prices, using an unknown sender, or breaking the format.

TRIAL B · OUTAGE DRILL

In a tabletop drill, the distributor "went dark" at 6 a.m. on a Friday. We timed how long each store took to place an emergency deli and dairy order, first without the binder and then, two weeks later, with it.

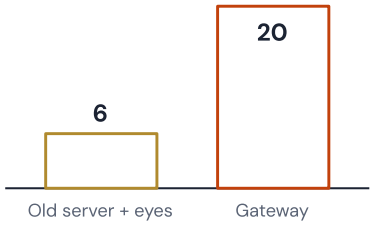
TRIAL C · CARD-SORT TRAINING

Thirty associates split into two groups of 15. One read a memo. The other used the Delivery Checkpoint Board and then followed the same files on the gateway dashboard. Everyone sorted 12 sample files, and passing meant 11 or more correct.

03 · WHAT HAPPENED

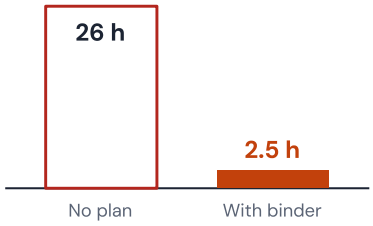
Every tampered file stopped at the door

Tampered files stopped (of 20)



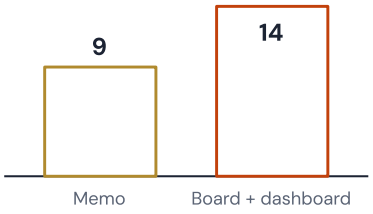
Fisher's exact, $p < .001$ · 1 false alarm in 20 good files

Hours to place an emergency perishable order



Descriptive · one drill per store

Passed the 12-file sort (of 15)



Fisher's exact, $p = .080$, a small sample

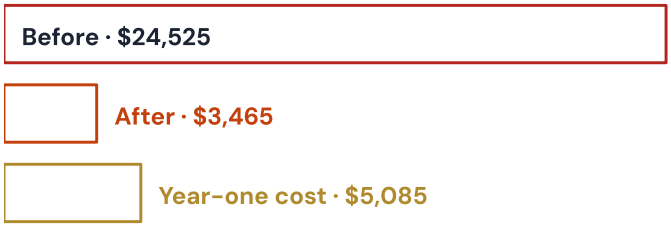
Illustrative data for a fictional grocer. Self-efficacy (1–5) rose from 2.6 to 4.0 with the board, and from 2.7 to 3.0 with the memo.

H1 is strongly supported, and the one false alarm took a two-minute phone call to clear. H2 shows the biggest practical gain: without a plan, it took more than a day to work out who to call, which for a deli means spoiled product. H3 shows a promising trend that needs a larger group to confirm.

04 · MONEY AND TIME

Small spend, big drop in expected loss

EXPECTED YEARLY LOSS (ALE)



Two risks: invoice fraud ($48,000 × 0.30 → 0.03) + 5-day outage

For an IT auditor, the key question is whether controls cost less than the loss they prevent (Gordon & Loeb, 2002). Building these checks into Glidecart's price-file import from the first release would also have avoided a later rebuild (Boehm, 1981).

3.1 : 1

return in year one: $15,975 net benefit on $5,085 spent

Managed file gateway	$2,400
Setup scripting (16 h)	$1,200
Associate training time	$1,080
Glidecart portal MFA setup	$300
3 Truck Stop Binders	$105
Total, year one	**$5,085**

Time saved: invoice checks drop from 12 to about 3 minutes, saving 36 hours a month, or about 430 hours a year.

A file that will not fit through the gate

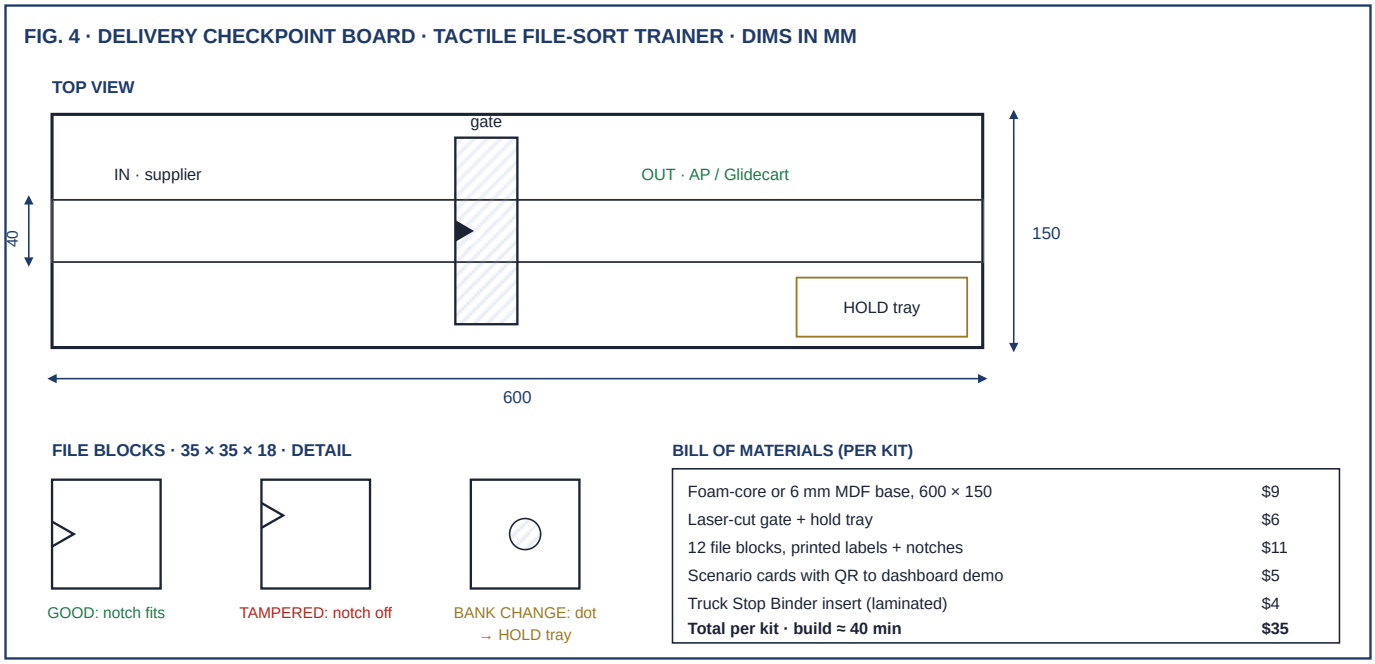


Figure 2. How the training kit is made. Each wooden "file" has a notch that stands for its digital fingerprint, and only a file with the matching notch slides through the gate. A tampered file's notch is shifted, so it physically will not pass. A raised dot marks a bank-detail change, and that file goes to the HOLD tray for a call-back. The parts cost $35, and it takes about 40 minutes to build.

THE PROBLEM TRAIL: ONE HABIT, FIVE STEPS

1

Notice

"Our dairy supplier just emailed new bank details."

2

Guess

"Is this a real change or a fake invoice?"

3

Test in software

Open the gateway dashboard and find the file in the "held" list, along with the reason it was held.

SOFTWARE

4

Prove by hand

Find the matching block. It has the raised dot, so it goes to HOLD.

TACTILE

5

Act

Call the supplier at the number in the binder, never the number in the email.

BUILDING SELF-EFFICACY ON PURPOSE

Store associates rarely think of themselves as part of security. Self-efficacy, the belief that "I can do this," changes that (Bandura, 1977), and workers who feel capable follow security practices more often (Rhee et al., 2009).

Mastery
The board gives a quick early success: the right block slides through.

Seeing others
Pairs watch each other sort before taking a turn.

Encouragement
The store manager praises good holds out loud.

Calm body
The board is a game, not a test, which keeps stress low.

WHY HANDS-ON LEARNING WORKS

The trail follows problem-based learning: a real problem first, then the tools to solve it (Hmelo-Silver, 2004). It also follows Kolb's (1984) cycle of doing, reflecting, and trying again. The tactile step helps workers who learn better by touch than by reading, and it turns an invisible idea, a file fingerprint, into something they can hold (Wilson, 2002).

> *"A plan no one has rehearsed is not a control."*
>
> TRIAL B · 26 HOURS WITHOUT THE BINDER, 2.5 WITH IT

TAKEAWAYS FOR AUDITORS

- List every internet-facing file-transfer service and its patch date. Replace self-hosted servers with managed services where you can.
- Test the file checks yourself: send a tampered test file and confirm it is held.
- Ask to see the offline fallback, and ask when it was last practiced.

LIMITATIONS

The grocer, Glidecart's role, and all data are fictional. The drill measured one Friday morning per store, and the card sort used 15 people per group. A real evaluation should repeat the drill each quarter and use real, de-identified file samples.

CONCLUSION

A small grocer cannot stop attacks on its distributor, but it can control the door its files come through and keep a plan for when the truck stops. Here, about $5,000 in the first year cut expected losses by more than $21,000, saved about 430 hours of AP time, and turned a 26-hour scramble into a 2.5-hour routine.

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